POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5014B | Semester V | Theory | 4 Credits

Remote Sensing and GIS

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5014B as Remote Sensing and GIS in Semester V for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Guwahati’s Remote Sensing and GIS course and NPTEL IIT Bombay’s course on Remote Sensing and GIS for rural development. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. GIS outputs, satellite interpretations and spatial analysis must be distinguished from authoritative legal boundary surveys, safety-critical infrastructure data or officially approved planning documents. Use verified software and current data sources.

Verified source note: Verified sources support remote sensing basics/corrections, digital image processing, classification, photogrammetry, thermal/microwave/high-resolution remote sensing, GIS fundamentals, spatial data structures and applications in agriculture, water, soil and disaster management. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Remote Sensing and GIS (Geographic Information Systems) studies the acquisition and analysis of spatial information from satellite and aerial platforms. It develops the ability to interpret imagery, process digital data, manage spatial information and apply geospatial technology to civil engineering challenges in water, agriculture, soil and disaster management while respecting the technical and legal boundaries of authoritative data.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□;
> □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□
> □□□□□□□□.

Prerequisite foundation

Basic surveying, engineering graphics, computer fundamentals, coordinate geometry, elementary physics and measurement awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain remote sensing fundamentals, electromagnetic radiation, sensors, platforms and satellite data corrections.
2. Apply digital image processing, enhancement and classification techniques to satellite imagery.
3. Explain photogrammetry, thermal, microwave and high-resolution remote sensing concepts.
4. Explain GIS fundamentals, spatial data structures, analysis methods and applications in rural and infrastructure development.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain remote sensing principles, EMR interactions, sensor/platform types and image correction basics.	Understand
CO2	Apply digital image processing, enhancement and classification to interpret satellite imagery.	Apply
CO3	Explain photogrammetry, thermal and microwave remote sensing and their specific application areas.	Understand
CO4	Explain GIS components, data structures, spatial analysis and applications in engineering and development.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Remote Sensing Fundamentals and Satellite Data	Remote sensing overview; electromagnetic radiation (EMR) and spectrum; interactions with atmosphere and earth features; sensors and platforms; satellite data acquisition and error correction basics.	Approx. 14 hours	CO1	Module 1
2	Digital Image Processing and Classification	Digital image fundamentals; image enhancement and filtering; image transformation; supervised and unsupervised classification; accuracy assessment and interpretation.	Approx. 16 hours	CO2	Module 2
3	Photogrammetry and Advanced Remote Sensing	Photogrammetry basics; aerial photography and scale; stereoscopy and DEM generation; thermal, microwave and high-resolution remote sensing; specific sensor applications.	Approx. 16 hours	CO3	Module 3
4	GIS Fundamentals and Geospatial Applications	GIS components and data structures (vector/raster); spatial data management; spatial analysis and queries; applications in water, agriculture, soil conservation and disaster management.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Remote Sensing Fundamentals and Satellite Data

Remote sensing overview; electromagnetic radiation (EMR) and spectrum; interactions with atmosphere and earth features; sensors and platforms; satellite data acquisition and error correction basics.

CO1 · Approx. 14 hours

Digital Image Processing and Classification

Digital image fundamentals; image enhancement and filtering; image transformation; supervised and unsupervised classification; accuracy assessment and interpretation.

CO2 · Approx. 16 hours

Photogrammetry and Advanced Remote Sensing

Photogrammetry basics; aerial photography and scale; stereoscopy and DEM generation; thermal, microwave and high-resolution remote sensing; specific sensor applications.

CO3 · Approx. 16 hours

GIS Fundamentals and Geospatial Applications

GIS components and data structures (vector/raster); spatial data management; spatial analysis and queries; applications in water, agriculture, soil conservation and disaster management.

CO4 · Approx. 14 hours

MODULE 1

Remote Sensing Fundamentals and Satellite Data

Remote sensing overview; electromagnetic radiation (EMR) and spectrum; interactions with atmosphere and earth features; sensors and platforms; satellite data acquisition and error correction basics.

Approx. 14 hours | CO1 | Understand · Apply

Learning goals

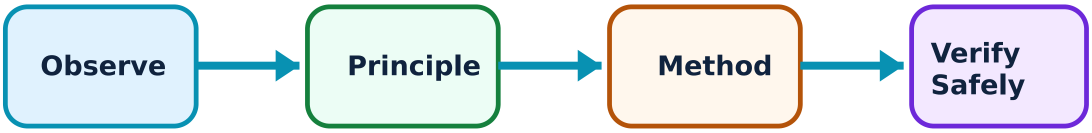
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Remote Sensing Fundamentals and Satellite Data: identify the system, state its principle, follow the correct method and verify the result safely.

1. Remote sensing principles and EMR−

Remote sensing acquires information about earth features without physical contact by measuring reflected or emitted electromagnetic radiation. The EMR spectrum, atmospheric windows and spectral signatures of water, soil and vegetation form the basis for interpreting satellite data.

Study and application method

Sketch the EMR spectrum and identify atmospheric windows; describe the spectral signature differences between healthy vegetation, dry soil and clear water.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the EMR spectrum and identify atmospheric windows; describe the spectral signature differences between healthy vegetation, dry soil and clear water.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Spectral signatures vary with moisture, health, time and sensor quality. Do not rely on visual appearance alone for critical classifications.

2. Sensors, platforms and data corrections−

Sensors (passive/active) on satellite or aerial platforms capture data with specific spatial, spectral, radiometric and temporal resolutions. Raw data require geometric and radiometric corrections to account for sensor errors, atmospheric effects and earth curvature before use.

Study and application method

Compare two sensor types (e.g., optical vs. microwave) and list the correction steps required before the data can be used for a mapping task.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare two sensor types (e.g., optical vs. microwave) and list the correction steps required before the data can be used for a mapping task.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application

□□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Uncorrected satellite data can lead to significant errors in distance, area and location. Use only verified, corrected data for engineering applications.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Digital Image Processing and Classification

Digital image fundamentals; image enhancement and filtering; image transformation; supervised and unsupervised classification; accuracy assessment and interpretation.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

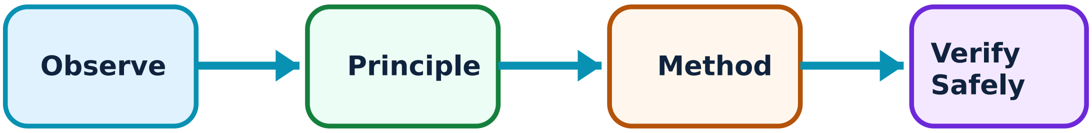
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Digital Image Processing and Classification: identify the system, state its principle, follow the correct method and verify the result safely.

1. Image enhancement and filtering−

Digital image processing improves imagery for human interpretation or machine analysis. Enhancement (contrast, brightness) and filtering (smoothing, sharpening) help isolate features, while transformations (like NDVI) can highlight specific characteristics like vegetation health.

Study and application method

Apply a conceptual enhancement workflow to an example image: identify noise, choose a filter, perform contrast stretching and explain the result.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Apply a conceptual enhancement workflow to an example image: identify noise, choose a filter, perform contrast stretching and explain the result.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Excessive filtering or enhancement can introduce artifacts or hide real features. Always document processing steps and compare results with original data.

2. Satellite image classification−

Classification groups pixels into land-cover categories. Supervised classification uses known 'training areas' to guide the algorithm, while unsupervised classification groups pixels by statistical similarity. Accuracy assessment (e.g., error matrix) verifies the result against ground truth.

Study and application method

Outline a classification plan: select training areas, choose an algorithm, generate a land-use map and perform a conceptual accuracy check.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Outline a classification plan: select training areas, choose an algorithm, generate a land-use map and perform a conceptual accuracy check.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Classification results are statistical estimates. They must be validated by ground-truthing and should not be used for legal or safety-critical decisions without verification.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Photogrammetry and Advanced Remote Sensing

Photogrammetry basics; aerial photography and scale; stereoscopy and DEM generation; thermal, microwave and high-resolution remote sensing; specific sensor applications.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

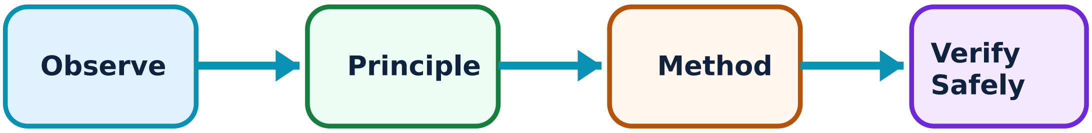
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Photogrammetry and Advanced Remote Sensing: identify the system, state its principle, follow the correct method and verify the result safely.

1. Photogrammetry and terrain modelling−

Photogrammetry uses overlapping photographs to determine the geometry and position of objects. Stereoscopic pairs allow for 3D viewing and the generation of Digital Elevation Models (DEMs), which are essential for slope, drainage and volume analysis.

Study and application method

Calculate the scale of an aerial photograph and describe the steps to generate a simple elevation profile from a stereoscopic pair.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the scale of an aerial photograph and describe the steps to generate a simple elevation profile from a stereoscopic pair.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Photogrammetric accuracy depends on flight height, camera quality and ground control points. Do not use unverified DEMs for hydraulic or structural design.

2. Thermal, microwave and high-resolution sensors−

Thermal sensors detect heat (emissivity); microwave (radar) sensors are active and can penetrate clouds/canopy; high-resolution sensors provide fine detail for urban and infrastructure mapping. Each has specific uses in moisture detection, disaster monitoring and detailed planning.

Study and application method

Select the appropriate advanced sensor for three scenarios (e.g., night-time monitoring, flood mapping in monsoon, urban road planning) and justify the choice.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Select the appropriate advanced sensor for three scenarios (e.g., night-time monitoring, flood mapping in monsoon, urban road planning) and justify the choice.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Advanced sensors require specialised processing and interpretation skills. Radar and thermal data can be complex to interpret correctly.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

GIS Fundamentals and Geospatial Applications

GIS components and data structures (vector/raster); spatial data management; spatial analysis and queries; applications in water, agriculture, soil conservation and disaster management.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

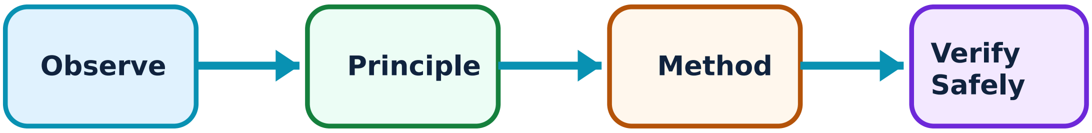
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for GIS Fundamentals and Geospatial Applications: identify the system, state its principle, follow the correct method and verify the result safely.

1. GIS data structures and analysis−

GIS organises spatial information into vector (points, lines, polygons) and raster (grid cells) formats. Spatial analysis uses queries, buffering, overlay and network analysis to answer geographic questions and support decision-making.

Study and application method

Build a conceptual GIS project for a road alignment: identify vector/raster layers, perform an overlay analysis to find constraints and run a query for suitable paths.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Build a conceptual GIS project for a road alignment: identify vector/raster layers, perform an overlay analysis to find constraints and run a query for suitable paths.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: GIS results reflect the quality of input data. Errors in digitising, attribute data or coordinate systems can lead to incorrect analysis results.

2. Geospatial applications in development−

Remote sensing and GIS support rural and infrastructure development by monitoring agriculture, managing water resources, planning soil conservation, assessing disaster risk and tracking development initiatives using tools like QGIS.

Study and application method

Map a geospatial workflow for a watershed management project: identify data sources, analysis steps, output maps and how the results inform field decisions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map a geospatial workflow for a watershed management project: identify data sources, analysis steps, output maps and how the results inform field decisions.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application

□□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: GIS maps are decision aids, not final plans. Field verification and stakeholder consultation are essential before implementing infrastructure or land-use changes.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Remote Sensing Fundamentals and Satellite Data

Use the concept flow to revise observation, principle, method and verification.

Module 2: Digital Image Processing and Classification

Use the concept flow to revise observation, principle, method and verification.

Module 3: Photogrammetry and Advanced Remote Sensing

Use the concept flow to revise observation, principle, method and verification.

Module 4: GIS Fundamentals and Geospatial Applications

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Sketch spectral signatures for four different earth features and explain the differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Outline the steps for supervised classification of a satellite image, including accuracy assessment.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate scale and displacement for a hypothetical aerial photograph.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Design a GIS overlay analysis to identify suitable sites for a small check dam.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Map a remote-sensing workflow for monitoring flood impact in a rural district.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Remote Sensing Fundamentals and Satellite Data

Explain Remote sensing principles and EMR and state one correct method, application or precaution.—

Remote sensing acquires information about earth features without physical contact by measuring reflected or emitted electromagnetic radiation. The EMR spectrum, atmospheric windows and spectral signatures of water, soil and vegetation form the basis for interpreting satellite data.

Answer framework: Sketch the EMR spectrum and identify atmospheric windows; describe the spectral signature differences between healthy vegetation, dry soil and clear water.

Important point: Spectral signatures vary with moisture, health, time and sensor quality. Do not rely on visual appearance alone for critical classifications.

Explain Sensors, platforms and data corrections and state one correct method, application or precaution.−

Sensors (passive/active) on satellite or aerial platforms capture data with specific spatial, spectral, radiometric and temporal resolutions. Raw data require geometric and radiometric corrections to account for sensor errors, atmospheric effects and earth curvature before use.

Answer framework: Compare two sensor types (e.g., optical vs. microwave) and list the correction steps required before the data can be used for a mapping task.

Important point: Uncorrected satellite data can lead to significant errors in distance, area and location. Use only verified, corrected data for engineering applications.

Module 2 — Digital Image Processing and Classification

Explain Image enhancement and filtering and state one correct method, application or precaution.−

Digital image processing improves imagery for human interpretation or machine analysis. Enhancement (contrast, brightness) and filtering (smoothing, sharpening) help isolate features, while transformations (like NDVI) can highlight specific characteristics like vegetation health.

Answer framework: Apply a conceptual enhancement workflow to an example image: identify noise, choose a filter, perform contrast stretching and explain the result.

Important point: Excessive filtering or enhancement can introduce artifacts or hide real features. Always document processing steps and compare results with original data.

Explain Satellite image classification and state one correct method, application or precaution.−

Classification groups pixels into land-cover categories. Supervised classification uses known 'training areas' to guide the algorithm, while unsupervised classification groups pixels by statistical similarity. Accuracy assessment (e.g., error matrix) verifies the result against ground truth.

Answer framework: Outline a classification plan: select training areas, choose an algorithm, generate a land-use map and perform a conceptual accuracy check.

Important point: Classification results are statistical estimates. They must be validated by ground-truthing and should not be used for legal or safety-critical decisions without verification.

Module 3 — Photogrammetry and Advanced Remote Sensing

Explain Photogrammetry and terrain modelling and state one correct method, application or precaution.−

Photogrammetry uses overlapping photographs to determine the geometry and position of objects. Stereoscopic pairs allow for 3D viewing and the generation of Digital Elevation Models (DEMs), which are essential for slope, drainage and volume analysis.

Answer framework: Calculate the scale of an aerial photograph and describe the steps to generate a simple elevation profile from a stereoscopic pair.

Important point: Photogrammetric accuracy depends on flight height, camera quality and ground control points. Do not use unverified DEMs for hydraulic or structural design.

Explain Thermal, microwave and high-resolution sensors and state one correct method, application or precaution.−

Thermal sensors detect heat (emissivity); microwave (radar) sensors are active and can penetrate clouds/canopy; high-resolution sensors provide fine detail for urban and infrastructure mapping. Each has specific uses in moisture detection, disaster monitoring and detailed planning.

Answer framework: Select the appropriate advanced sensor for three scenarios (e.g., night-time monitoring, flood mapping in monsoon, urban road planning) and justify the choice.

Important point: Advanced sensors require specialised processing and interpretation skills. Radar and thermal data can be complex to interpret correctly.

Module 4 — GIS Fundamentals and Geospatial Applications

Explain GIS data structures and analysis and state one correct method, application or precaution.−

GIS organises spatial information into vector (points, lines, polygons) and raster (grid cells) formats. Spatial analysis uses queries, buffering, overlay and network analysis to answer geographic questions and support decision-making.

Answer framework: Build a conceptual GIS project for a road alignment: identify vector/raster layers, perform an overlay analysis to find constraints and run a query for suitable paths.

Important point: GIS results reflect the quality of input data. Errors in digitising, attribute data or coordinate systems can lead to incorrect analysis results.

Explain Geospatial applications in development and state one correct method,

application or precaution.−

Remote sensing and GIS support rural and infrastructure development by monitoring agriculture, managing water resources, planning soil conservation, assessing disaster risk and tracking development initiatives using tools like QGIS.

Answer framework: Map a geospatial workflow for a watershed management project: identify data sources, analysis steps, output maps and how the results inform field decisions.

Important point: GIS maps are decision aids, not final plans. Field verification and stakeholder consultation are essential before implementing infrastructure or land-use changes.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Remote Sensing Fundamentals and Satellite Data.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Digital Image Processing and Classification.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Photogrammetry and Advanced Remote Sensing.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: GIS Fundamentals and Geospatial Applications.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Remote sensing overview; electromagnetic radiation (EMR) and spectrum; interactions with atmosphere and earth features; sensors and platforms; satellite data acquisition and error correction basics..

2. Develop a complete answer or practical plan for: Digital image fundamentals; image enhancement and filtering; image transformation; supervised and unsupervised classification; accuracy assessment and interpretation..
3. Develop a complete answer or practical plan for: Photogrammetry basics; aerial photography and scale; stereoscopy and DEM generation; thermal, microwave and high-resolution remote sensing; specific sensor applications..
4. Develop a complete answer or practical plan for: GIS components and data structures (vector/raster); spatial data management; spatial analysis and queries; applications in water, agriculture, soil conservation and disaster management..

LAST-MILE REVISION

Quick Revision

Module 1: Remote Sensing Fundamentals and Satellite Data

Remote sensing overview; electromagnetic radiation (EMR) and spectrum; interactions with atmosphere and earth features; sensors and platforms; satellite data acquisition and error correction basics.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Digital Image Processing and Classification

Digital image fundamentals; image enhancement and filtering; image transformation; supervised and unsupervised classification; accuracy assessment and interpretation.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Photogrammetry and Advanced Remote Sensing

Photogrammetry basics; aerial photography and scale; stereoscopy and DEM generation; thermal, microwave and high-resolution remote sensing; specific sensor applications.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: GIS Fundamentals and Geospatial Applications

GIS components and data structures (vector/raster); spatial data management; spatial analysis and queries; applications in water, agriculture, soil conservation and disaster management.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Remote sensing principles and EMR	Remote sensing acquires information about earth features without physical contact by measuring reflected or emitted electromagnetic radiation.
Sensors, platforms and data corrections	Sensors (passive/active) on satellite or aerial platforms capture data with specific spatial, spectral, radiometric and temporal resolutions.
Image enhancement and filtering	Digital image processing improves imagery for human interpretation or machine analysis.
Satellite image classification	Classification groups pixels into land-cover categories.
Photogrammetry and terrain modelling	Photogrammetry uses overlapping photographs to determine the geometry and position of objects.
Thermal, microwave and high-resolution sensors	Thermal sensors detect heat (emissivity); microwave (radar) sensors are active and can penetrate clouds/canopy; high-resolution sensors provide fine detail for urban and infrastructure mapping.
GIS data structures and analysis	GIS organises spatial information into vector (points, lines, polygons) and raster (grid cells) formats.
Geospatial applications in development	Remote sensing and GIS support rural and infrastructure development by monitoring agriculture, managing water resources, planning soil conservation, assessing disaster risk and tracking development initiatives using tools like QGIS.

LEARNING RESOURCES

References

Recommended remote-sensing and GIS references

1. B. Bhatta, Remote Sensing and GIS.
2. Lillesand, Kiefer, and Chipman, Remote Sensing and Image Interpretation.
3. Kang-tsung Chang, Introduction to Geographic Information Systems.
4. P. A. Burrough and R. A. McDonnell, Principles of Geographical Information Systems.

Approved public remote-sensing and GIS sources

- https://nptel.ac.in/courses/105103193
- https://onlinecourses.nptel.ac.in/noc26_ce15/preview

These sources provide the verified study basis while official Course 5014B detail is unavailable; they do not replace official surveying standards, land-record rules, safety-critical data requirements or authorised geospatial procedures.